

JEE MAIN REVISION MASTER: TRIGONOMETRIC EQUATIONS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 3 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Standard	Trigonometric Functions And Equations — Standard Calculations & Problem Archetypes	2 PYQs	Quantitative Analysis
Standard	Trigonometric Functions And Equations — Core Principles & Governing Laws	1 PYQs	Core Principle
Standard	Trigonometric Functions And Equations — Applied Cases, Reactions & Traps	0 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Trigonometric Functions And Equations — Standard Calculations & Problem Archetypes	$y=\tan x$, $\cos^2 x + \sin^2 x = 1$, $\sin^2 x + \sin^4 x$
Trigonometric Functions And Equations — Core Principles & Governing Laws	$\cos x = \sin x $

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Trigonometric Functions And Equations — Standard Calculations & Problem Archetypes (Standard Yield — 2 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Trigonometric Functions And Equations.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2025 Shift] Q#1 (MCQ): "If $\cos x + \cos^2 x = 1$, then the value of $\sin^2 x + \sin^4 x$ is..." [Ans: N/A]
- [2022 Shift] Q#2 (MCQ): "The trigonometric function $y = \tan x$ in the II quadrant..." [Ans: N/A]

3.2 Trigonometric Functions And Equations — Core Principles & Governing Laws (Standard Yield — 1 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Trigonometric Functions And Equations.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Trigonometric Functions And Equations.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Trigonometric Functions And Equations PYQs.

Representative Source PYQs:

- [2019 Shift] Q#3 (MCQ): "If $\cos x = |\sin x|$ then, the general solution is..." [Ans: N/A]

3.3 Trigonometric Functions And Equations — Applied Cases, Reactions & Traps (Standard Yield — 0 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Trigonometric Functions And Equations.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2025 Shift] Q#1 (MCQ): "If $\cos x + \cos^2 x = 1$, then the value of $\sin^2 x + \sin^4 x$ is..." [Ans: N/A]
- [2022 Shift] Q#2 (MCQ): "The trigonometric function $y = \tan x$ in the II quadrant..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2025 (Q#1)

If $\cos x + \cos^2 x = 1$, then the value of $\sin^2 x + \sin^4 x$ is

- (A) -1
- (B) 1
- (C) 0
- (D) 2

Correct Answer: None

Solution Analysis: $\cos x + \cos^2 x = 1 \Rightarrow 1 - \cos^2 x = \cos x \Rightarrow \sin^2 x = \cos x \Rightarrow \sin^2 x + \sin^4 x \Rightarrow \cos x + (\cos x)^2 = 1$

Benchmark Problem 2: KCET 2019 (Q#3)

If $\cos x = |\sin x|$ then, the general solution is

- (A) $x = n\pi + (-1)^n(\pi/4), n \in \mathbb{Z}$
- (B) $x = n\pi \pm (\pi/4), n \in \mathbb{Z}$
- (C) $x = (2n+1)\pi \pm (\pi/4), n \in \mathbb{Z}$
- (D) $x = 2n\pi \pm (\pi/4), n \in \mathbb{Z}$

Correct Answer: None

Solution Analysis: We have, $|\sin x| = \cos x \Rightarrow \sin^2 x = \cos^2 x \Rightarrow 2\cos^2 x = 1 \Rightarrow \cos x = \frac{1}{\sqrt{2}}$ (since, $\cos x$ can not be negative) $\Rightarrow x = 2n\pi \pm (\pi/4)$